

TIC 29495157

Eclipsing binary candidate dossier - Star Vetter

PERIOD ALIAS

BLS EPHEMERIS

Period 5.147449 d (ALIASSED, see below)
-> corrected period: 10.294897 d
Epoch (t0) 1328.1391 BTJD
Duration 0.1700 d
Depth 8.579 % (85,786 ppm)
SNR 333.2

CATALOG CROSS-MATCH (novelty)

VSX	no match
ASAS-SN	no match
Gaia eclipsing-binary table	no match
Gaia variable classification	no match
TESS EB catalog (Prsa+2022)	no match
ExoFOP TOI list	no match

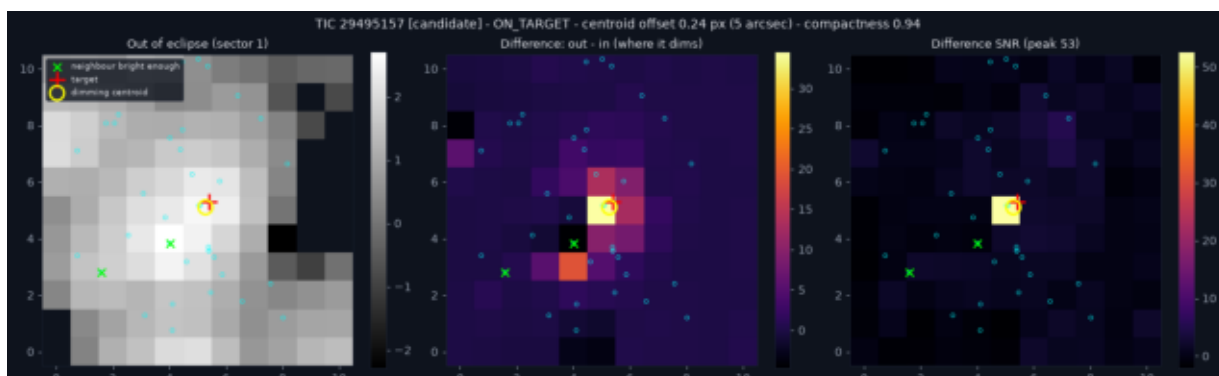
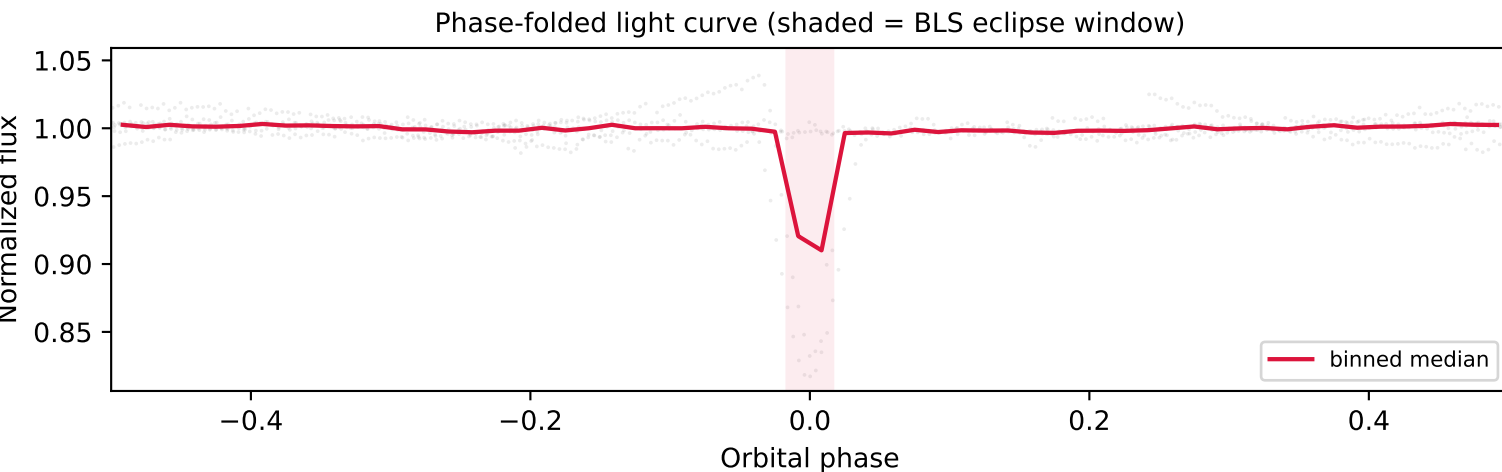
INDEPENDENT CHECKS (f3i gates)

Eclipses observed 4 (gate ≥ 3)
Robust depth ratio 1.05 (gate ≥ 0.50)
Robust depth sigma 86.0 (gate ≥ 7.0)
Odd/even z-score 20.17 (ALIAS FLAGGED)
Secondary eclipse not detected

Verdict: NOVEL

PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)

Sector 1 light-curve dip in this sector: 94.0 sigma t0 refined by 0.0000 phase
Difference-image peak SNR 52.62 (gate ≥ 5.0) Compactness 0.94 (gate ≥ 0.50)
Dimming centroid offset 0.24 px (5 arcsec) (gate ≤ 1.0 px, i.e. on the target)
Nearest bright-enough neighbour in cutout: TIC 29495166
1.79 px away, dTmag -0.48 - farther from the dimming than the target



Caveat (PERIOD ALIAS): Odd/even eclipse-depth z-score is 20.17 (gate > 3.0): alternating eclipses differ significantly in depth, the signature of a true period double what BLS found. Pixel-check confirmation and eclipse timing are unaffected - report the corrected period, 10.294897 d, in any write-up.